

Humidification
Final Lab Report
Unit Operation Lab II
Date: 13 Feb 2020

Commented [DL1]: 89/100

Team 3, Thursday, PM section
Chiu, Jon
Djonlich, Melissa C
Li, Yier
Slykas, Cheryl Lynn
Tristano, Caleb Francis



Commented [ZA2]: Interesting image!

Table of Contents

Executive Summary	12
Introduction.....	23
Theory	34
Result.....	89
Discussion.....	1413
Reference	1514
Appendix I: Data Tabulation/Graphs.....	1615
Appendix II: Error Analysis.....	1918
Appendix III: Sample Calculations.....	2019
Appendix IV: Individual Team Contributions.....	2524

Executive Summary

The mass and heat transfer coefficients were experimentally determined for a wetted-wall humidification unit. Prior to wetting the column walls, the temperature and humidity of the incoming air was measured by using an electronic hygrometer and wet and dry bulb thermometers. After the column walls were wetted and reached steady state, the air humidity and the temperature of air and water at various locations within the column were measured via hygrometer, thermometers, and thermocouples. Measurements were made for three different flowrates: 3, 7, and 10 L/min. Then the incoming air was heated to two different temperatures via heat exchanger. The measurements were repeated with the above three air flowrates for both heated air temperatures. The incoming dry air humidity was found to be 8.1%. The temperatures of the water in the column for the air flowrates of 3, 7, and 10 L/min ranged between 75 and 88 F. The humidity of the air for each flowrate was found to be 51.2, 59.5, and 63.6% respectively. The temperatures of the water in the column for the heated air flowrates of 3, 7, and 10 L/min ranged between 85 and 94 F. The humidity of the air for each flowrate was found to be 71.1, 53.5, and 36.6% for medium temperature air, 73.2, 68.7, and 58% for high temperature air. The mass and heat transfer coefficients were calculated as 0.352-1.304 lb/ft²s and 0.0526 to 0.6412 BTU/ft² F s respectively.

Commented [DL3]: 8/10

Commented [DL4]: Flowrates of which substance, air or water?

Commented [DL5]: L/min or SCFM?

Commented [ZA6]: Relative humidity

Commented [ZA7]: give specific temperature

Commented [DL8]: You should present the trends in this experiment. What are the other parameters change? Such as the dimensionless groups and the NtG.

Introduction

Humidification units are used in the chemical engineering industry to study the effects of both mass and heat transfer in a process. The results of studying a humidification or dehumidification unit can be applied to other processes that have rates effected by both mass and heat transfer. One such unit is a chemical reactor. A chemical reactor can have heat transfer from a chemical reaction, as well as, mass transfer [3].

The objective of this experiment is to obtain a set of data from which the mass and heat transfer coefficients can be calculated. The resulting mass and heat transfer coefficients will be plotted as a function of the Reynold number. The Sherwood equation divided by $Sc^{0.44}$ will also be plotted as a function of the Reynolds number on a log-log plot. These plots will be used to make a comparison between the results of this experiment against known values for air and water systems in wetted-wall columns. The relationship between the measured data and the calculated mass and heat transfer coefficients will be analyzed and compared against the expected results. The HTU and NTU will also be used for further comparison of results with the expected values.

The wetted-wall humidification unit in this experiment consists of a Plexiglas column with air and water feed connections. The water is pumped into the top of the column and is measured by a rotameter. A weir distributes the water to the sides of the column. The air comes from the bottom of the column and is supplied by the house supply and is also measured by a rotameter. The air and water flow counter-currently.

The temperature of the water is measured in the column via three thermocouples. The temperature of the incoming water is measured in the holding tank also via thermocouple. The incoming air temperature is measured by thermocouple and the outgoing air is measured by wet-bulb thermometer or a hygrometer. The humidity of the air is also determined by hygrometer. A heat exchanger is able to heat the incoming air, which is used to collect data for various humidification rates.

Commented [DL9]: 10/10

Commented [DL10]: Good. But rearrange the citation numbers.

Commented [DL11]: Good.

Commented [DL12]: Good.

Theory

The use of a wet bulb thermometer is often used to measure humidity. Wet bulb temperature is measured by having the air flow come into contact with the thermometer. The thermometer has a bulb that is covered by a wick saturated with water. When the wet and dry bulb temperatures are known, a psychrometric chart can be used to determine the humidity.[1]. Humidification is the transfer of water from the liquid phase into a gaseous mix of air and water vapor. [2] This involves heat and mass transfer between a liquid water and air. Heat and mass transport are usually described in terms of transport coefficients using the equations for heat and mass flux as shown below in equation 1 and 2 respectively. [3]

$$q = h \Delta T \quad (1)$$

$$N_A = K_c \Delta c_A \quad (2)$$

Where,

q = heat flux (energy/(area*time))

N_A = mass flux (moles/(area*time))

h = heat transfer coefficient

K_c = mass transfer coefficient

ΔT = difference in temperature (driving force)

Δc_A = difference in concentration (driving force)

The driving forces are the temperature and concentration differences and are linear in these equations. [3]

A wetted-wall column or falling film column is widely used in laboratories to study heat and mass transport of water into air. Air flows countercurrent to a falling film of water down the side a column of certain height and diameter. The process occurs adiabatically, and the water is cycled through the column. [3] The temperature and concentration profiles in terms of humidity are illustrated at the water-gas interface. See Figure 1. [2]

Commented [DL13]: 9/10

Units and assumptions. Note that some of the empirical equations only valid in a specific unit system, like CGS, or FPS.

Commented [DL14]: What's the assumptions using Newton's law and Fick's law?

Commented [DL15]: Units.

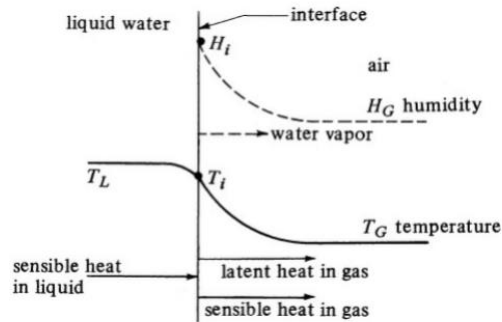


Figure 1. Temperature and concentration profile in upper part of cooling tower [2]

Water vapor diffuses from the interface to the bulk gas phase with a driving force in the gas phase of $(H_i - H_G)$, where H is the humidity and is defined as the kg of water/kg dry air. Since water is pure liquid, there is no driving force in the liquid phase. The temperature driving force in the liquid is $T_L - T_i$ where T_L is the liquid temperature and T_i is the temperature at the interface. The temperature driving force in the gas phase is $T_i - T_G$ where T_G is the bulk gas temperature. Heat flows from the bulk liquid to the interface and from the interface to the bulk gas phase. Latent heat also diffuses into the gas phase from the interface in the water vapor. [2]

Figure 1 represents the upper portion of a cooling tower, where the lower part of the tower will have the bulk water temperature higher than the wet bulb temperature of the air. [2]

Dimensionless groups are often used to obtain numerical values for the transfer coefficients. Experimental data is obtained using the wetted-wall column to obtain these values. [2]

$$Sh = 0.023 * Re^{0.83} * Sc^{0.44} \quad (3)$$

$$Sh = k_c * \frac{L}{D_{AB}} \quad (4)$$

Where,

Sc = Schmidt number

Re = Reynolds number in terms of pipe diameter and relates laminar or turbulent flow of water

Sh = Sherwood number

k_c = mass transfer coefficient for concentration driving force

L = length of tower

D_{AB} = diffusion coefficient for fluid where mass transport occurs. (the air)

Figure 2 illustrates a wetted-wall column where the dotted box illustrates a cross-section of the column. A mass balance is made inside the box. See equation 5.[3]

$$d(Gy) = k_c c (y_i - y) dA \quad (5)$$

Where,

G = total gas flow rate (moles/time)

y = mol fraction of water in gas

y_i = mole fraction of water at interface

c = molar concentration of the gas

dA = Area of differential column height

k_c = mass transfer coefficient

$d(Gy)$ = differential change in moles of water in gas over differential column height

Z = differential section of height of column

P = perimeter of column.

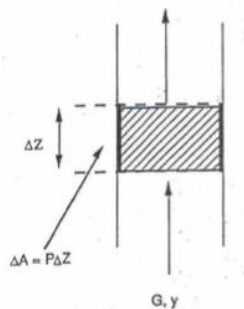


Figure 2. Cross-sectional area of wetted-wall column [3]

This equation is only valid when the transferring species is at low concentration. The bulk flow term must be included in the expression if the transferring species is in high concentration. For humidification, the concentration of water vapor in the air is defined in equation 6. [3]

$$Y' = \left\{ \frac{y}{(1-y)} \right\} \left\{ \frac{\text{molecular weight of water}}{\text{molecular weight of air}} \right\} \quad (6)$$

Commented [DL16]: Good.

Where,

$Y' = \text{concentration}$ of water vapor in air

$y = \text{mol fraction}$ of water

dG_y can then be written as equation 7.[3]

$$d(Gy) = \left\{ \frac{G_s'}{\text{molecular weight of air}} \right\} d(Y') \quad (7)$$

Where,

$G_s' = \text{mass flow rate of dry air}$

Equation 7 can then be applied to equation 5 to become equation 8. [3]

$$G_s' d(Y') = k_y (Y_i' - Y') P dz \quad (8)$$

Where,

$k_y = \text{the mass transport coefficient for mass ratio driving force and can be calculated with equation 9. [3]}$

$$k_y = k_c (\text{molecular weight of air}) c (1 - y)_M \quad (9)$$

Where,

$$(1 - y)_M = \frac{(y_1 - y_2)}{\ln \frac{(1 - y)_2}{(1 - y)_1}} \quad (10)$$

Where the numbers 1 and 2 denote the inlet and outlet compositions of the column, respectively. [3]

The box in Figure 2 can also be used to make an energy balance. Equation 11 describes the heat balance of the column. [3]

$$d(GC_p T) = h(T_i - T) dA \quad (11)$$

Where,

$C_p = \text{is the molar heat capacity of the gas}$

$T_i = \text{the temperature at the interface}$

$T = \text{temperature of the of the gas}$

Equation 11 can be written in terms of the mass flowrate of dry air, G_s' , and the humid heat of capacity of a gas C_s' . See Equation 12. [3]

$$G_s' C_s' dT = h(T_i - T) dA \quad (12)$$

Where,

$$C_s' = C_{Air}' + Y' C_{H_2O}'$$

Where,

C_{Air}' = the heat capacity of dry air

C_{H_2O}' = the heat capacity of water vapor.

Y' = vapor concentration of the gas.

For wetted-wall columns with recirculated water, Y' and T_i are constant and since C_s' is a function of Y' , it can also be assumed constant. [2] The resulting equation for humidification problems for mass and energy are equations 13 and 14.

$$\ln \frac{(Y_i' - Y_1')}{(Y_i' - Y_2')} = \frac{k_Y P z}{G_s} \quad (13)$$

$$\ln \frac{(T_i' - T_1')}{(T_i' - T_2')} = \frac{h P z}{G_s C_s} \quad (14)$$

Where 1 and 2 denote the inlet and outlet compositions and temperatures.

The number of transfer units (NTU) is equal to $\ln \frac{(Y_i' - Y_1')}{(Y_i' - Y_2')}$. It is the “measure of the number of discrete equilibrium stages necessary to achieve mass transfer. The height of transfer unit (HTU) describes the column height necessary to achieve mass transfer and the efficiency of contacting.” [3]

Result

Commented [DL17]: 15/15

Figure 3 compares the air flow rates and percent humidity in the wetted-wall column for the unheated air, medium temperature air, and high temperature air. For all temperatures, the humidity decreases as the flow rates increase. Unheated air and high temperature air seem to have a parallel graph with similar slopes. The humidity seems to decrease at an increasing rate as the flow rate increases for each temperature. The medium temperature air followed a more extreme humidity drop as flow rate increased.

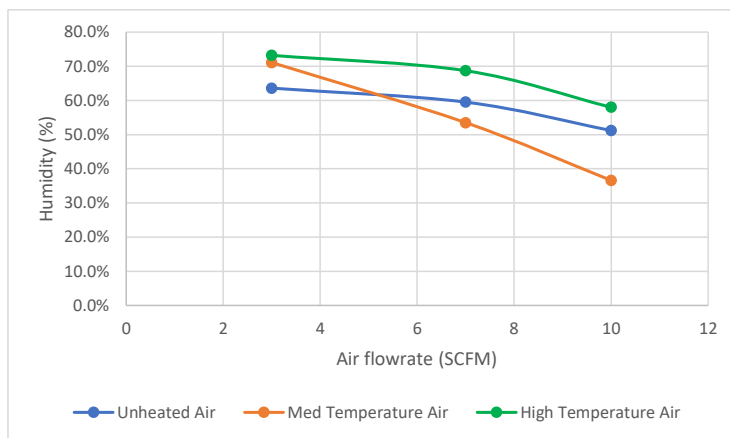


Figure 3 Air Flowrate vs Humidity

Figure 4 compares the Reynolds number (Re) transport coefficient for mass ratio driving force (k_y). At a Re of 1750, the transport coefficients are about the same for all air temperatures. As Re increases, the transport coefficients increase. The unheated air had the largest transport coefficients followed by the medium temperature air and the high temperature air. The relationship appears linear for all three temperatures.

Formatted: Not Superscript/ Subscript

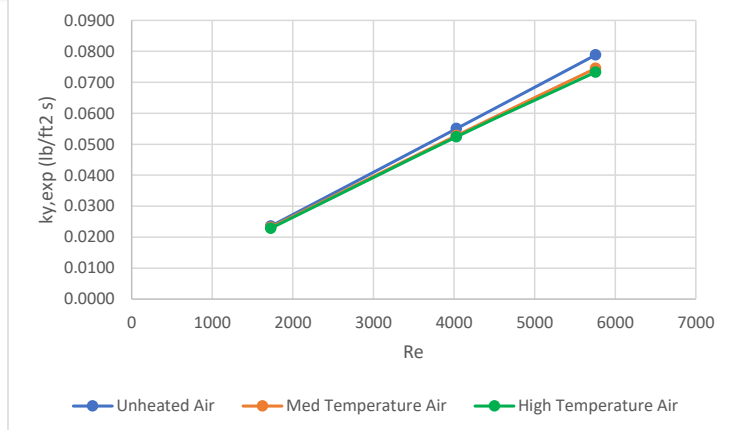


Figure 4 $k_{y,exp}$ vs Reynolds number

Figure 5 illustrates the relationship between the mass transfer coefficient with the concentration driving force vs Re. Again, as in Figure 4, as Re increases, k_c increases. k_c has the largest values as Re increases for medium heat air, followed by unheated air. There was little to no change in k_c values for high temperature air as Re increases.

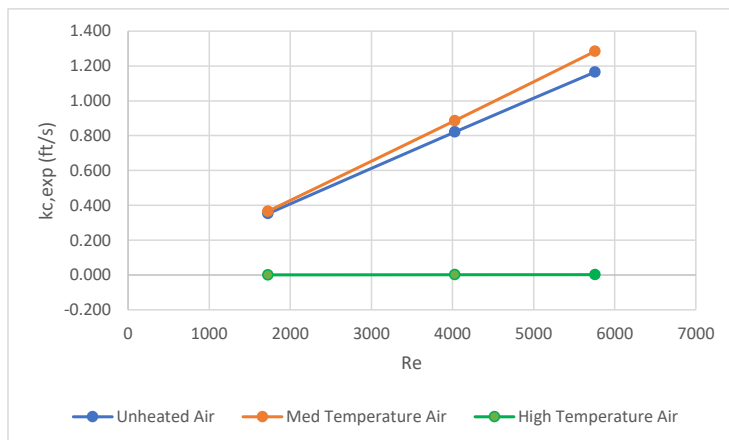


Figure 5 $k_{c,exp}$ Vs. Reynold Number

Commented [DL18]: So much difference between the high temperature data and others.

Figure 6 is the heat transfer coefficient vs the Re. The heat transfer coefficient is largest for the high temperature air, then medium temperature air and then unheated air. The heat transfer coefficient trends to increase as Re increase. The rate of increase seems to be linear in all cases and about the same for the medium and high temperature air. The unheated air heat transfer coefficient does increase as Re increases, but at a slower rate than the heated air.

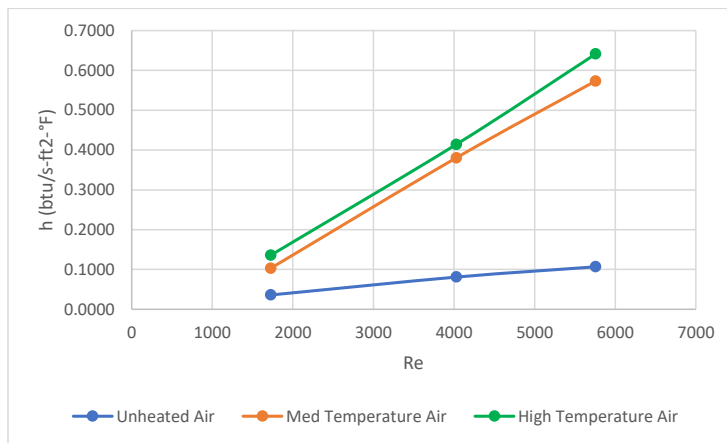


Figure 6 Heat Transfer Coefficient Vs. Reynold Number

The Sherwood number/Schmidt number ratio is related to the Re on a log-log plot. Experimental data is blue is compared to the literature. The two lines are parallel to each other and follow the same slopes, however, the experimental data is larger than the literature values.

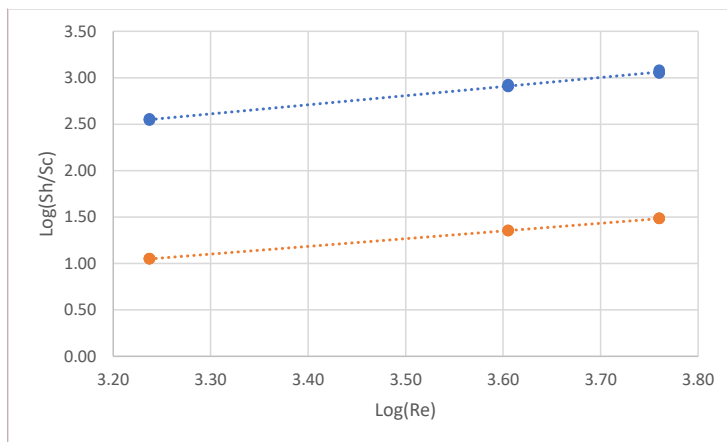


Figure 7 Sh/Sc Ratio vs Reynold Number in a log-log graph
 (Experimental data Top, Correlation Data Bottom)

Commented [DL19]: Log plots require the axes to be logarithm, but not the data. (But it's OK!)

Commented [DL20]: R² and fit functions are needed.

Figure 8 shows the relationship between flow rate of air and the number of transfer units. For the medium and high temperature air, the NTU increased as the flow rate increased. This was not the case for the unheated air, which decreased. The NTU is largest for high heat air, then medium heat, air and then unheated air.

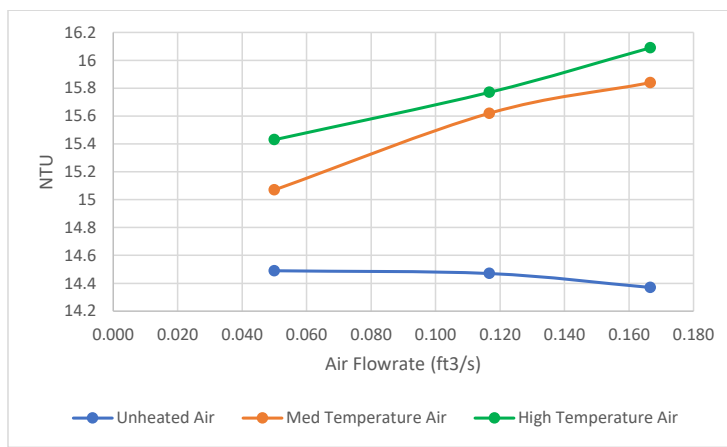


Figure 8 NTU vs Air Flowrate

Figure 9 shows how the height of transfer unit relates to flow rate and temperature of air. As the flow rate increased, the HTU decreased for the medium and high temperature air but increased for the unheated air. This is opposite from Figure 8 where the HTU is largest for unheated air and smallest for high heat air.

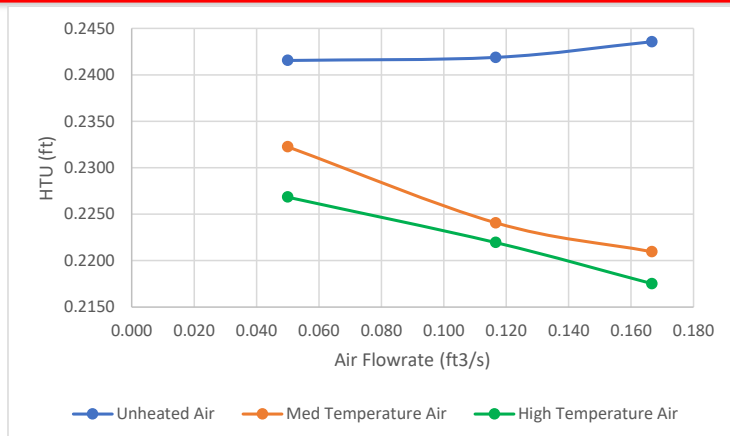


Figure 9 HTU vs Air Flowrate

Discussion

Looking at figure 3, we can see that the volumetric flowrate of the air has a strong effect on the humidity of the unit. The high temperature and unheated air showed about the same level of change in humidity with respect to air flowrate, and the medium temperature air showed the highest level of change in humidity with respect to air. Figure 4 shows that the mass transport coefficient changes significantly as Reynolds number (and therefore flowrate) changes, and that it doesn't change very much with respect to temperature. This is to be expected, as we expect effects from convective transport to be much larger than those from molecular/diffusive transport. Additionally, at higher air flowrates, evaporated water is removed from the column more quickly, keeping the concentration difference, and therefore the driving force for molecular transport high.

Figure 6 shows that the heat transfer coefficient increases with increasing flow, as well as increases with temperature. With respect to temperature, the increase seems to diminish, because it increases significantly between room temperature and medium temperature, but only increases a small amount between medium and high temperature. So, unlike the mass transport coefficient, the heat transfer coefficient appears to have a much stronger effect due to temperature differences. We expect this because temperature difference is the driving force for heat transfer.

By using our data and psychrometric charts and the diffusivity of water through air, we were able to calculate the Schmidt Number. Then that in addition to the Sherwood Number we used to calculate the ratio of the Schmidt to Sherman Numbers raised to the power of 0.44 was plotted against the Reynolds Number on a logarithmic scale to experimentally determine the pipe correlation from equation 4.

Commented [DL21]: 13/15

Commented [DL22]: Use third-person statement.

Commented [DL23]: Good discussion. References?

Commented [DL24]: Good discussion. But you should explain why the humidity is higher when the flow rate is lower, not only the change in humidity.

Commented [DL25]: Good discussion.

Commented [DL26]: Yes, and you should compare the theoretical and the experimental data, then give a discussion on why there're differences.

Reference

- [1] Green, Don W., Perry, Robert H. Perrys Chemical Engineers' Handbook. 8th Edition. McGraw-Hill. New York. 2008.
- [2] Geankoplis, Christie John, Hersel, A. Allen, Lepek, Daniel H. Transport Processes and Separation Process Principles. 5th Edition. Prentice Hall. Boston, 2018.
- [3] Zdunek, A. ChE 382: Unit Operations Laboratory: Humidification. Revision: Spring 2018v1.

Commented [DL27]: 5/10

You should cite more than 4 references NOT including the lab manual.

Commented [DL28]: Pages.

Appendix I: Data Tabulation/Graphs

Commented [DL29]: 10/10
Good.

Table 1 Operation Conditions

Parameter	Value	Unit
Gas Temperature	75	°F
Gas Pressure	14.7	psia
Relative Density of Air	0.0624	lb/ft ³
Column Diameter	0.187	ft
Column Height	3.5	ft
Column Perimeter	0.587	ft
Column Cross-Section Area	0.0275	ft ²
Diffusion Coefficient of Air to Water Vapor	2.96E-04	ft ² /s
Kinematic Viscosity of Air	1.66E-04	ft ² /s
Dynamic Viscosity of Air	1.23E-05	lbm/(ft-s)

Table 2 Collected Data

Flowrate	Air Only	Unheated Air			Med Temperature Air			High Temperature Air		
Water (GPM)	0	0.098	0.098	0.098	0.098	0.098	0.098	0.098	0.098	0.098
Air (SCFM)	5.0	3.0	7.0	10.0	3.0	7.0	10.0	3.0	7.0	10
T1 Wet Bulb (°F)	59	75	74	71	84	89	91	89	91	94
T2 Dry Bulb (°F)	75	79	79	77	96	115	122	106	119	129
T3 Column Top (°F)	75	88	87	84	92	91	90	94	93	91
T4 Mid Column (°F)	75	86	83	79	89	87	86	90	89	87
T5 Column Bottom (°F)	75	84	81	76	88	86	85	90	89	87
T6 Reservoir (°F)	75	83	79	75	87	86	85	90	88	87
T7 Incoming Air (°F)	74	70	72	71	121	145	150	137	152	160
Humidity (%)	8.1	63.6	59.5	51.2	71.1	53.5	36.6	73.2	68.7	58.0

Table 3 Psychrometric Properties

Run No.	Temperature (°F)		Humidity	Specific Volume (ft ³ /lb)	Enthalpy (Btu/lb)	Humidity Ratio (lb/lb)	Relative Humidity	Dew Point (°F)	Vapor Pressure (psi)	Saturated Vapor Pressure (psi)
	Wet	Dry								
1	59	75	8.1%	14.14	26.2	0.0075	39.2%	48.5	0.17	0.43
2	75	79	63.6%	14.49	39.3	0.0185	83.6%	73.6	0.41	0.49
3	74	79	59.5%	14.47	38.3	0.0177	79.7%	72.2	0.39	0.49
4	71	77	51.2%	14.37	35.5	0.0156	75.4%	68.6	0.35	0.46
5	84	96	71.1%	15.07	49.1	0.0237	61.7%	80.8	0.52	0.84
6	89	115	53.5%	15.62	55.6	0.0252	37.5%	82.7	0.55	1.47
7	91	122	36.6%	15.84	58.5	0.0262	32.0%	83.8	0.57	1.79
8	89	106	73.2%	15.43	55.7	0.0273	52.4%	85.0	0.60	1.14
9	91	119	68.7%	15.77	58.5	0.0269	35.6%	84.5	0.59	1.65
10	94	129	58.0%	16.09	63.0	0.0287	28.8%	86.5	0.62	2.16

Table 4 Dimensional Correlation

Run No.	Air Flow (ft ³ /s)	Velocity (ft/s)	Reynold Number (Re)	Schmidt Number (Sc)	Sherwood Number (Sh)		$\frac{Sh}{Sc^{0.44}}$		Log(Re)	$\log\left(\frac{Sh}{Sc^{0.44}}\right)$	
					Exp	Corr	Exp	Corr		Exp	Corr
1	0.083	3.03	2878	0.56	469	13.25	605	17	3.46	2.78	1.23
2	0.050	1.82	1727		279	8.67	359	11	3.24	2.56	1.05
3	0.117	4.25	4030		651	17.52	839	23	3.61	2.92	1.35
4	0.167	6.07	5757		932	23.56	1202	30	3.76	3.08	1.48
5	0.050	1.82	1727		274	8.67	354	11	3.24	2.55	1.05
6	0.117	4.25	4030		625	17.52	805	23	3.61	2.91	1.35
7	0.167	6.07	5757		881	23.56	1136	30	3.76	3.06	1.48
8	0.050	1.82	1727		271	8.67	349	11	3.24	2.54	1.05
9	0.117	4.25	4030		620	17.52	799	23	3.61	2.90	1.35
10	0.167	6.07	5757		867	23.56	1118	30	3.76	3.05	1.48

Table 5 Mass Transfer Coefficients

Run No.	Inlet Partial Pressure (y_{Ai})	Outlet Partial Pressure (y_A)	$(1 - y_{Ai})$	$(1 - y_A)$	$(1 - y_A)_{lm}$	$Conc$ $\left(\frac{lbmol}{ft^3}\right)$	$k_{y,exp}$ $\left(\frac{lb}{ft^2 \cdot s}\right)$	$k_{y,corr}$ $\left(\frac{ft}{s}\right)$	$k_{c,exp}$ $\left(\frac{lb}{ft^2 s}\right)$	$kc, corr$ $\left(\frac{ft}{s}\right)$
1	0.0116	0.0293	0.9884	0.9707	0.9796	0.0024	0.0397	7.76E-05	0.573	1.12E-03
2	0.0279	0.0333	0.9721	0.9667	0.9694	0.0024	0.0236	4.91E-05	0.352	7.34E-04
3	0.0266	0.0333	0.9734	0.9667	0.9700	0.0024	0.0550	9.93E-05	0.821	1.48E-03
4	0.0236	0.0313	0.9764	0.9687	0.9726	0.0024	0.0788	1.35E-04	1.165	1.99E-03
5	0.0353	0.0572	0.9647	0.9428	0.9537	0.0023	0.0232	4.64E-05	0.366	7.34E-04
6	0.0376	0.1000	0.9624	0.9000	0.9309	0.0022	0.0528	8.83E-05	0.886	1.48E-03
7	0.0390	0.1218	0.9610	0.8782	0.9190	0.0022	0.0745	1.16E-04	1.284	1.99E-03
8	0.0405	0.0776	0.9595	0.9224	0.9408	0.0022	0.0229	4.47E-05	0.375	7.34E-04
9	0.0399	0.1123	0.9601	0.8877	0.9234	0.0022	0.0524	8.68E-05	0.895	1.48E-03
10	0.0425	0.1470	0.9575	0.8530	0.9042	0.0021	0.0733	1.12E-04	1.304	1.99E-03

Table 6 Heat Transfer Coefficients with HTU and NTU

Run No.	$\Delta T1$ (°F)	$\Delta T2$ (°F)	LMTD (°F)	Heat Transfer Coefficient (BTU/ft ² °F s)	NTU	HTU (ft)
1	16	1	5.41	0.0526	14.14	0.2475
2	4	9	6.17	0.0360	14.49	0.2415
3	5	7	5.94	0.0809	14.47	0.2419
4	6	5	5.48	0.1067	14.37	0.2436
5	12	25	17.71	0.1034	15.07	0.2322
6	26	30	27.95	0.3806	15.62	0.2241
7	31	28	29.47	0.5734	15.84	0.2210
8	17	31	23.30	0.1360	15.43	0.2268
9	28	33	30.43	0.4144	15.77	0.2219
10	35	31	32.96	0.6412	16.09	0.2175

Appendix II: Error Analysis

Throughout the humidification experiment, there are few systematic errors and random errors. Several apparatuses may have errors in reading data. First, at collecting data, Humidifier Hygrometer is an error in taking a particular wet blub value because the airflow and humidity keep jumping around, and it is constantly dripping. To synthesize the time that the wet blub column is reached to steady-state condition, data kept at the same value, stay in five seconds were used as parameters. Due to this reason, the humidity of the wet-bulb steady-state, it is hard to determine. Secondly, the inlet airflow rate can be another systematic source of error, and reading is between 0.5 SCFM; therefore, the airflow rate can contribute some of the difference in experience result. The inlet airflow changes are close to the 3, 7 and 11 SCFM, Thurs, whose measured values should not exceed the experimental data by a large percentage. Moreover, the uncertainty of the water flow meter had random error fluctuate at 0.326 GPM. The digital thermometer is increased by 0.1 degrees at the time.

To observe the effect of a supplied low, medium, and hot stream on temperature, the digital thermometer was very slow and continuous increasing after 10 minutes, which might indicate that the heat steam supply had not delivered to a steady-state condition. At inlet airflow temperature Channel 7, the temperature gauge that was measuring the stream temperature was used as air inlet temperature, which can occur heat loss during the experiment at a different measurement.

Table 7 Uncertainty of The Apparatus

Uncertainty Chart		
Equipment List	Uncertainty	Unit
Temperature Gauge	± 1	°F
Water Flow Rotameter	± 0.01	GPM
Air Flow Rotameter	± 0.5	SCFM
Thermometer	± 1	°F
Hygrometer	± 0.1	% Humidity

Commented [DL30]: 9/10

Quantitative error analysis about the dimensionless groups' correlation is needed.

Commented [DL31]: 3, 7, 11 or 3, 7, 10?

Appendix III: Sample Calculations

Commented [DL32]: 10/10

Using Data from Run No. 1 with air at 5 SCFM and water at 0 GPM

- Convert the air flowrate (q_{air}) to ft^3/s .

$$q_{air} = 5 \frac{ft^3}{min} \times \frac{1 min}{60 s} = \mathbf{0.083 \frac{ft^3}{s}}$$

- Calculate the velocity of air.

$$v = \frac{q_{air}}{A} = \frac{4q_{air}}{\pi d^2}$$

Where

$$v = \text{velocity} \left(\frac{ft}{s} \right)$$

$$A = \text{cross-sectional area of the column} (ft^2)$$

$$d = \text{column diameter } 2.25 (in)$$

Sample Calculation:

$$v = \frac{4 \left(0.083 \frac{ft^3}{s} \right)}{\pi \left(2.25 \text{ inch} \times \frac{1 ft}{12 \text{ inch}} \right)^2} = \mathbf{0.34 \frac{ft}{s}}$$

- Calculate the column inner perimeter (P).

$$P = \pi d = (0.187 ft) \pi = \mathbf{0.587 ft}$$

- Calculate the Reynolds number.

$$Re = \frac{\rho v d}{\mu}$$

Where

$$Re = \text{Reynolds number}$$

$$\rho = \text{density of air} \left(\frac{lb}{ft^3} \right)$$

$$\mu = \text{dynamic viscosity of air} \left(\frac{lbm}{ft^2 * s} \right)$$

Sample Calculation:

$$Re = \frac{0.0624 \frac{lb}{ft^3} * 0.187 ft * 3.03 \frac{ft}{s}}{1.23 * 10^{-5} \frac{lbm}{ft * s}} = \mathbf{2878}$$

- Calculate Schmidt number.

$$Sc = \frac{V}{D_{AB}}$$

Where

$$V = \text{Kinematic viscosity of air} \left(\frac{ft^2}{s} \right)$$

$$D_{AB} = \text{Diffusion coefficient of air} \left(\frac{ft^2}{s} \right)$$

Sample Calculation:

$$Sc = \frac{1.66 \times 10^{-4} \frac{ft^2}{s}}{2.96 \times 10^{-4} \frac{ft^2}{s}} = \mathbf{0.56}$$

- Calculate the experimental Sherwood number.

$$Sh_{exp} = \frac{k_{Y,exp} l}{D_{AB}}$$

Where

$$Sh_{exp} = \text{experimental Sherwood number}$$

$$k_{Y,exp} = \text{experimental mass transfer coefficient} \left(\frac{ft}{s} \right)$$

$$l = \text{column length or height (ft)}$$

Sample Calculation:

$$Sh_{exp} = \frac{0.0397 \frac{ft}{s} * (3.5 ft)}{2.98 \times 10^{-4} \frac{ft^2}{s}} = \mathbf{469}$$

- Calculate the correlation Sherwood number (Sh_{corr}).

$$Sh_{corr} = 0.023 * Re^{0.83} Sc^{0.44} = 0.023 * (2878^{0.83})(0.56^{0.44}) = \mathbf{13.25}$$

- Calculate the Sherwood / Schmidt Ratio.

$$\frac{Sh_{exp}}{Sc^{0.44}} = \frac{243}{0.55^{0.44}} = \mathbf{316}$$

$$\frac{Sh_{corr}}{Sc^{0.44}} = \frac{13.25}{0.56^{0.44}} = \mathbf{17.19}$$

- Calculate the logarithm of the Reynolds number.

$$\log(Re) = \log(1727) = \mathbf{3.24}$$

- Calculate the logarithm of the Sherwood / Schmidt Ratio.

$$\log\left(\frac{Sh_{exp}}{Sc^{0.44}}\right) = \log(316) = \mathbf{2.50}$$

$$\log\left(\frac{Sh_{corr}}{Sc^{0.44}}\right) = \log(11.19) = \mathbf{1.05}$$

- Calculate the log mean average composition of air over the column length.

$$(1 - y_A)_{lm} = \frac{(1 - y_A) - (1 - y_{Ai})}{\ln\left(\frac{1 - y_A}{1 - y_{Ai}}\right)}$$

Where

$$(1 - y_A)_{lm} = \text{log mean average composition of air} \left(\frac{lb}{ft^3}\right)$$

$$(1 - y_{Ai}) = \text{composition at the inlet of the tube}$$

$$(1 - y_A) = \text{composition at the outlet of the tube}$$

Sample Calculation:

$$(1 - y_A)_{lm} = \frac{0.9707 - 0.9884}{\ln\left(\frac{0.9707}{0.9884}\right)} = \mathbf{0.9796 \frac{lb}{ft^3}}$$

- Calculate the experimental mass transfer coefficient.

$$k_{Y,exp} = \frac{(1 - y_A)_{lm} * q_{air}}{P * L}$$

Where

P = Column inner perimeter (ft^2)

L = Length of the Column (ft)

$k_{Y,exp}$ = experimental mass transfer coefficient ($\frac{ft}{s}$)

Sample Calculation:

$$k_{Y,exp} = \frac{0.9796 \frac{lb}{ft^3} * 0.083 \frac{ft^3}{s}}{0.587 ft * 3.5 ft} = 0.0397 \frac{lb}{ft^2 s}$$

- Calculate the experimental mass transfer coefficient ($k_{c,exp}$).

$$k_{c,exp} = \frac{k_{Y,exp}}{MW_{air} C (1 - y)_M}$$

$$k_{c,exp} = \frac{0.0397 \frac{lb}{ft^2 \cdot s}}{28.96 \frac{lb}{lbmol} \left(0.0024 \frac{lbmol}{ft^3} \right) (0.9796)} = 0.573 \frac{ft}{s}$$

- Calculate the correlation mass transfer coefficient.

$$k_{Y,corr} = k_{c,corr} MW_{air} C (1 - y_A)_{lm}$$

Where

$k_{Y,corr}$ = correlation mass transfer coefficient ($\frac{lb}{ft^2 \cdot s}$)

MW_{air} = molecular weight of air ($\frac{lb}{lbmol}$)

C = molar concentration ($\frac{lbmol}{ft^3}$)

Sample Calculation:

$$k_{Y,corr} = \left(1.12 * 10^{-3} \frac{ft}{s} \right) \left(28.97 \frac{lb}{lbmol} \right) \left(2.44 * 10^{-3} \frac{lbmol}{ft^3} \right) (0.9796)$$

$$= 7.76 * 10^{-5} \frac{lb}{ft^2 s}$$

- Calculate the correlation mass transfer coefficient ($k_{c,corr}$).

$$k_{c,corr} = \frac{Sh_{corr} D_{AB}}{l}$$

$$k_{c,corr} = \frac{13.25 \left(2.98 * 10^{-4} \frac{ft^2}{s} \right)}{3.5 ft} = 1.12 * 10^{-3} \frac{ft}{s}$$

- Calculate the heat transfer coefficient.

$$h = \frac{q_{air} C_p}{Pl} LMTD$$

Where

$$h = \text{heat transfer coefficient} \left(\frac{Btu}{ft^2 \text{ } ^\circ F s} \right)$$

$$C_p = \text{heat capacity of air} \left(\frac{Btu}{lb \text{ } ^\circ F} \right)$$

LMTD = Log Mean Temperature Difference ($^\circ F$)

ΔT_1 = Interface – Inlet temperature ($^\circ F$)

ΔT_2 = Interface – Outlet temperature ($^\circ F$)

Sample Calculation:

$$h = \frac{0.083 \frac{ft^3}{s} * 0.240 \frac{Btu}{lb \text{ } ^\circ F}}{1.178 ft * 3.5 ft} * \left[\frac{16 - 1}{\ln \left(\frac{16}{1} \right)} \right] = 0.0526 \frac{Btu}{ft^2 \text{ } ^\circ F s}$$

- Calculate the number of transfer units (NTU).

$$NTU = \ln \left(\frac{Y'_i - Y'_1}{Y'_i - Y'_2} \right) = \frac{k_{c,exp} Pl}{q_{air}} = \frac{0.577 * 1.178 ft * 3.5 ft}{0.083 \frac{ft^3}{s}} = 14.14$$

- Calculate the height of transfer unit (HTU).

$$HTU = \frac{l}{NTU} = \frac{3.5 ft}{14.14} = 0.2475 ft$$

Appendix IV: Individual Team Contributions

NAME: Chiu, Jon	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present both days
Pre-Lab Editing	.5	Format layouts
Experimental Objective (Prelab)		
Apparatus (prelab)	2	Write up
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	.5	Graphs
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	4	Write up
Error Analysis (final lab)		
Sample Calculations (final Lab)	2	Write up
Power Point Presentation		
Total		

NAME: Djonlich, Melissa	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present in both days
Pre-Lab Editing	1	editing
Experimental Objective (Prelab)		
Apparatus (prelab)	1	drafted
Materials & Supplies (prelab)		
Procedure (prelab)	0.5	drafted
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)	2	drafted
Introduction (Final lab)	2	drafted
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation		

Total	13.5	
NAME: Li, Yier	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present in both days
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)	1	Wrote up
Materials & Supplies (prelab)	0.5	Wrote up
Procedure (prelab)		
Project Safety Evaluation (prelab)	0.5	Wrote up, MSDS
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)	2	Wrote up
Sample Calculations (final Lab)		
Power Point Presentation	2.5	Wrote up
Total	13.5	

NAME: Slykas, Cheryl	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	
Pre-Lab Editing	2	
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)	2	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	1	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)	5	
Results (final lab)	2	
Discussion (final lab)		
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	
Total	22	

NAME: Tristano, Caleb	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)		
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation		
Total		



Commented [DL33]: Cool picture.